

Printing, Variables & Input

When we have a coding language, we want it to be able to spit something out. If a program calculates the tax; we want to get it back obviously.

- **PRINT** is a *command*, used to make Python spit out text.
- The format for that command goes; `print("What I want to say")`

There's no need for any other code; this alone will make the system spit out the text "What I want to say"

On the Top right there's a RUN icon; ▶ Click it

- It's gonna open something at the bottom
 - And it'll say "What I want to say"

`What I want to say`

You don't only have to print out just text, you can do math in it too.

- Say you; want to get your tax `print(11 * 1.13)`

Not having the `"""` lets you give the print command something other than text like numbers for math, you can COMBINE math and text too actually.

- This returns the product; and text. `print(11 * 1.13, "is your tax.")`

`12.43 is your tax.`

If you want to print with numbers or other without commas; look into f strings later.

This alone is pretty boring right? You can do the same in a calculator, what you can't do is save "Variables".

- Variables allow you to save ANYTHING; say, instead of having to write your address anytime your going home; you can just say "Home". You can use that anywhere, and if you need to change it, "Home" is still home, but the actual address changed.

```
variable_name = "Something (Or don't put the quotes for numbers)"  
print(variable_name)
```

Let's say you have something that prints the price + tax for every product in your store. But, one day the tax rate changes; if all of your calculations used a Variable, you can update one to update all.

So, that'd look like this; (Variables, also should exist OUTSIDE the "")

```
product_a_name = "Eggs"  
product_a_price = 11  
tax_rate = 0.13  
print(product_a_price * tax_rate + product_a_price, "is your total cost  
for", product_a_name)
```

12.43 is your total cost for Eggs

All we've really done is made a glorified calculator with text Python is way more than just math. Right now, we have to MANUALLY go into the code to edit; Imagine if you had to go into your calculator app and find some line of code to give it your numbers to calculate.

- So, what we can do is let us INPUT what we want into a variable that it can use. From the same place that we see our output in.

The Input is done by the "input" command, and we save that into a variable. If we don't save it, we can't put it anywhere; it'd be useless.

```
our_variable = input("Some question: ")
```

For our case, how about we ask the USER their product and price of that product.

```
your_product = input("What is your product? ")  
your_product_price = input("What is the price of your product? ")  
tax_rate = 0.13  
print(your_product_price * tax_rate + your_product_price, "is your total  
cost for", your_product)
```

When we RUN this; we can see we can TYPE at the bottom there.

- Let's say BANANA, then hit enter. And let's say its 14.50 dollars; hit ent

What is your product? Banana
What is the price of your product? 14

And, an ERROR popped up, Why?

This is because of something called “Value Types”.

- The input command normally assumes that what you’re inputting is a word (Called a STRING). *And you can’t do math with a WORD.*
 - So we want to convert our INPUT to a number “Value Type”.
 - This isn’t magic, if we put in 14, *this will tell Python.* I want the number 14; **Not the Text 14.**
 - It won’t magically turn something like “Pineapple” into a number, if it’s something that is supposed to be a number; *it’ll convert to one.*

Without changing our old code, we can put the type we want before our input.

```
your_product_price = float(input("What is the price of your product? "))
```

- This “float” is actually a command which takes in the VALUE from the input command, and tries to make it into a decimal value type.
 - So 14.50 goes from text, AKA a string; to a decimal AKA a float.

Now we can do the math correctly.

```
What is your product? Banana
What is the price of your product? 14.5
16.385 is your total cost for Banana
```

The next lab will start to go over LISTS; *and how we can use lists to actually do less work after the program is actually running (Automation)*